

Guide to SQLAlchemy

Description

- SQLAlchemy is a package that creates a bridge between a SQL server and Python.
- It uses a `dialect` which is a specific type of driver that allows you to specify which particular SQL server SQLAlchemy has to build a bridge with so Python can understand.

Main Dialects

1. PostgreSQL
 - Uses a driver called `psycopg2`
2. Microsoft SQL
 - Uses a driver called `pymssql`
3. MySQL
 - Uses a driver called `pymysql`

Installation and Setup

1. Install the following packages:
 - `requirements.txt`

```
pandas
sqlalchemy
psycopg2-binary
```

```
pip install -r requirements.txt
```

Example - Connect to PostgreSQL

- Basics of a connection string

```
postgresql://username:password@ip_address:port/database
```

- Insecure Python example

```
username = ''  
password = ''  
host = ''  
port = ''  
database = ''
```

```
connection_string = f'postgresql://{username}:{password}@{host}:{port}/{database}'
```

- Secure Python example

1. Load all the different components as environment variables.

- Create a file called `.env`

```
POSTGRES_USER="something"  
POSTGRES_PASSWORD="something"  
POSTGRES_HOST="something"  
POSTGRES_PORT="something"  
POSTGRES_DATABASE="something"
```

- Install the `python-dotenv` package so you can load your environment variables.

```
pip install python-dotenv
```

```
from os import getcwd, environ
from dotenv import load_dotenv

env_path = getcwd() + "/.env"

# Load environment variables
load_dotenv(env_path)

# Credentials
host = environ.get('POSTGRES_HOST')
user = environ.get('POSTGRES_USER')
password = environ.get('POSTGRES_PASSWORD')
port = environ.get('POSTGRES_PORT')
database = environ.get('POSTGRES_DATABASE')

connection_string = f'postgresql://{username}:{password}@{host}:{port}/{database}'
```

Build the Connection

```
from sqlalchemy import create_engine

# Variable
engine = create_engine(connection_string)

# Turn the engine into a Connection
cursor = engine.connect()

# Do whatever you want with the cursor
```

Read Data into Pandas

```
import pandas as pd

# Create a table output as a DataFrame
df = pd.read_sql_table('users', con=cursor, schema='public')

# Results of the DataFrame
df.info()

# SQL Query Results
query = """

SELECT *
FROM schema.table_name
WHERE col_name = 'Active'

"""

df = pd.read_sql_query(query=query, con=cursor)
```